

REPRO Report

Target Dummies crash the game when saved in editor

Case md-12579-terra-overnight-01 | Mindustry | Awaiting human

Revision: f7ededb3a1950c396e5c113d96a19dd88e71da9a

Reported platform: Windows (reported); Linux (planned evaluation) | Build: Unspecified

Generated 2026-09-14 19:14 UTC from the current saved case.

Validation is incomplete or failed. Inspect the recorded gates.

Reproduction	Time to first proof	Replay actions
5/5 clean runs	8:12	15 to 12

Player report

Platforms

Windows

Build

bleeding-edge build 27733

Issue

If a Target Dummy is placed in the editor and the map is saved, the game crashes. The map also causes an error if it is attempted to be loaded afterwards.

Steps to reproduce

1. Place a target dummy in the editor
2. Save the map

Mods used

None

Save file

(As described above, this map is unloadable in-game.)
[dummy.zip](https://github.com/user-attachments/files/31756506/dummy.zip)

(Crash) logs

[crash-report-09_02_2026_16_06_14.txt](https://github.com/user-attachments/files/31756477/crash-report-09_02_2026_16_06_14.txt)

Expected behavior: Saving an editor map containing a Target Dummy should complete without crashing, and the saved map should load without an error.

Initial questions

Recorded at intake; later findings and validation may address these questions.

- Whether build 27733 exactly corresponds to the supplied target commit.
- Exact Windows version, architecture, graphics configuration, and Java/runtime details.
- Whether the issue occurs with a new blank map or depends on map settings, rules, content, or prior editor actions.
- Whether the crash occurs immediately on save, during save completion, or after a particular save-related action (for example, save dialog confirmation or overwrite).
- Exact error shown when loading the saved map.
- Whether the attached map archive contains the exact map produced by the reported reproduction steps.
- Whether the crash is reproducible on Linux; Linux is listed as planned evaluation, not as a reported affected platform.
- Whether Target Dummy refers to a built-in unit/block/content item and its exact internal content identifier.
- Whether the save was a new map save or an overwrite of an existing map.
- Whether any non-mod local configuration, custom content, or altered game files were present.

Recorded reproduction

1. Open Editor from main menu
2. Create a new editor map
3. Focus the modal map name input underline
4. Input new blank map name
5. Confirm creation of blank map
6. Switch editor palette to block placement
7. Focus editor block search
8. Search for Target Dummy block
9. Select Target Dummy block from results
10. Place selected Target Dummy on blank map
11. Open editor map menu to save
12. Save map containing Target Dummy

Source diagnosis

TargetDummy.TargetDummyBuild.write dereferences unitTeam without a null check (`write.i(unitTeam.id)`). `unitTeam` is declared without initialization and is only defaulted to the building's team in `updateTile`; consequently, a newly placed dummy whose update/defaulting has not occurred can throw `NullPointerException` while map serialization invokes its write method. This exactly matches the supplied oracle frame at TargetDummy.java:271. The failed write can leave the just-written map incomplete, which is a supported explanation for the reported later load error, but the exact load error was not supplied.

`core/src/mindustry/world/blocks/defense/TargetDummy.java`

This is the direct crash site named by the empirical oracle and contains a concrete null dereference consistent with the newly placed editor building state. It is the primary localization; no future fix is inferred.

`core/src/mindustry/io/SaveVersion.java`

This is the verified caller/path that turns the Target Dummy state defect into an editor save crash. It does not itself create the null value.

core/src/minindustry/io/SaveVersion.java

This is the likely path for the reported post-crash load error, but without the produced map or exact load stack trace it cannot be established whether the file is partial, or what read failure occurs.

core/src/minindustry/io/MapIO.java

Relevant error propagation boundary for editor map saving, but no evidence indicates it mishandles the exception or causes the invalid state.

Proposed patch

Why this patch should work

Cause: `unitTeam` is declared without an initial value (line 91) and is normally defaulted from the building team only in `updateTile()` (line 107). `write()` can run before that update, but line 271 unconditionally reads `unitTeam.id`, causing the reported null-pointer crash while saving a newly placed Target Dummy.

The changed serialization line writes the building's `team.id` when `unitTeam` has not yet been initialized; otherwise it writes the explicitly selected dummy team exactly as before. The building team is the same default `updateTile()` would assign, so the saved value preserves the intended default and lets the map save complete rather than leaving a partial output.

This is a one-line, local serialization guard. It keeps the existing save format (one integer in the same position), so no map format/version change is required. Validation should run the existing replay and also verify both a newly placed default dummy and a dummy configured to another team save and reload with their expected teams.

Risks and tradeoffs

- This relies on `team` being initialized for a building being serialized. That is the existing default source used in `updateTile()` and should hold for placed map buildings, but malformed manually constructed objects are not addressed.
- The patch prevents this specific write-time null dereference; it does not establish the exact cause of any already-corrupted map load failure. Previously interrupted/truncated map files may remain unreadable and should be regenerated.
- Other pre-update code paths that dereference `unitTeam` (for example configuration UI) are intentionally unchanged because the confirmed regression is the serialization crash.

Code changes

Lines starting with - are removed; lines starting with + are added. Unmarked lines provide context.

```
diff --git a/core/src/minindustry/world/blocks/defense/TargetDummy.java b/core/src/minindustry/world/blocks/defense/TargetDummy.java
index be2cd52..030221f 100644
--- a/core/src/minindustry/world/blocks/defense/TargetDummy.java
+++ b/core/src/minindustry/world/blocks/defense/TargetDummy.java
@@ -268,7 +268,7 @@ public class TargetDummy extends Block{
    write.f(unitArmor);
    write.f(resetTime);
    write.f(dummySize);
-   write.i(unitTeam.id);
+   write.i((unitTeam == null ? team : unitTeam).id);
}

@Override
```

Additional fix checks

Each candidate validation run first executes the unchanged recorded trigger, then these additional checks. Planning experiments are separate from fresh validation repetitions.

1. Observe saved editor state after frozen trigger
2. Open editor map menu for return navigation
3. Quit editor to saved-map list
4. Confirm exit from saved editor
5. Open editor map selection
6. Open saved DummySaveTest map
7. Load saved map into editor
8. Zoom in on retained map objects without editing
9. Increase normal editor zoom around retained object
10. Activate non-modifying block picker
11. Sample retained object with non-modifying picker
12. Observe stable reloaded editor state with sampled object

Validation results

Validation is incomplete or has failed. The recorded results below determine what has been verified.

Check	Result	Recorded evidence
Regression before patch	PASS	Trigger observed in 5/5 fresh baseline runs.
Candidate build	PASS	Build exit code 0
Existing tests	FAIL	Test exit code 1. Failing tests: :tests:test::ModTestAllure.begin(). Network-enabled candidate failures cannot be waived by an offline baseline.
Original replay after patch	PASS	5/5 reached expected state without the symptom; bug seen 0 times. Each replay included the unchanged trigger and 12 additional fix-check actions.
Smoke test	PASS	Clean desktop launch and live process after startup; deeper gameplay smoke coverage is not implemented.

Source analysis limitations

These are the source reviewer's recorded notes. That stage inspects code; see Recorded reproduction and Validation results for executed game checks.

- No crash report, full stack trace, generated map artifact, or exact load-error text was available to inspect.
- The supplied source does establish the crash-site null dereference and serializer call chain, but does not by itself prove the exact editor scheduling condition that leaves `unitTeam` null in every reproduction.
- The subsequent load failure is localized only as a probable consequence of interrupted/corrupt output; its specific parser failure remains unverified.
- Build 27733-to-commit correspondence and Windows-specific runtime details were not independently verified.

Usage and review

Each replay starts with a fresh profile. Reproduction counts describe this case; action reduction is bounded and is not proof of a globally minimal replay.

107 model calls; 962,177 input tokens; 42,432 output tokens. Usage accumulates across jobs on this case. PDF export uses saved data and makes no model calls.

The proposed patch lives in a disposable game workspace. Handoff approval records a local review decision and does not publish or merge changes upstream. Visual checks are model-based; startup smoke does not establish broad gameplay coverage.

Investigation timeline

All timestamps are UTC. First and latest entries are saved event times, not active work durations. Repeated attempts and pauses are retained. Selected milestones appear below; the workspace has the full searchable log.

Triage / 4 records

First entry: 14 Sep 2026 15:13:40 UTC
Latest entry: 14 Sep 2026 15:54:45 UTC

14 Sep 2026 15:13:40 | Report received. Ready for triage.

14 Sep 2026 15:54:45 | Reported crash when saving an editor map containing a Target Dummy; the saved map is also reported to produce an error when loaded afterward.

Reproduce / 203 records

First entry: 14 Sep 2026 15:54:45 UTC
Latest entry: 14 Sep 2026 16:41:35 UTC

14 Sep 2026 15:54:45 | Fetching the exact revision and preparing build dependencies.

14 Sep 2026 16:41:35 | Symptom observed. Observed game process state: {'pid': 45, 'exit_code': 1, 'started_at': 1789404063.9871168, 'running': False, 'launched': True}. Required log signature: 'TargetDummy\$TargetDummyBuild.write(TargetDummy.java:271)'.

Reduce / 170 records

First entry: 14 Sep 2026 16:05:44 UTC
Latest entry: 14 Sep 2026 16:13:14 UTC

14 Sep 2026 16:05:44 | Proposing action deletions and accepting them only after fresh baseline replays.

14 Sep 2026 16:13:14 | Verified in 5/5 replays; 15 12 actions.

Localize / 48 records

First entry: 14 Sep 2026 16:13:14 UTC
Latest entry: 14 Sep 2026 16:42:18 UTC

14 Sep 2026 16:13:14 | Tracing observed behavior into the pre-fix source.

14 Sep 2026 16:42:18 | Saving the confirmed replay as an executable regression test.

Validate / 462 records

First entry: 14 Sep 2026 16:42:18 UTC
Latest entry: 14 Sep 2026 18:49:42 UTC

14 Sep 2026 16:42:18 | Designing a small candidate diff in the disposable repository.

14 Sep 2026 18:49:42 | Expected state reached; symptom absent.

Review / 2 records

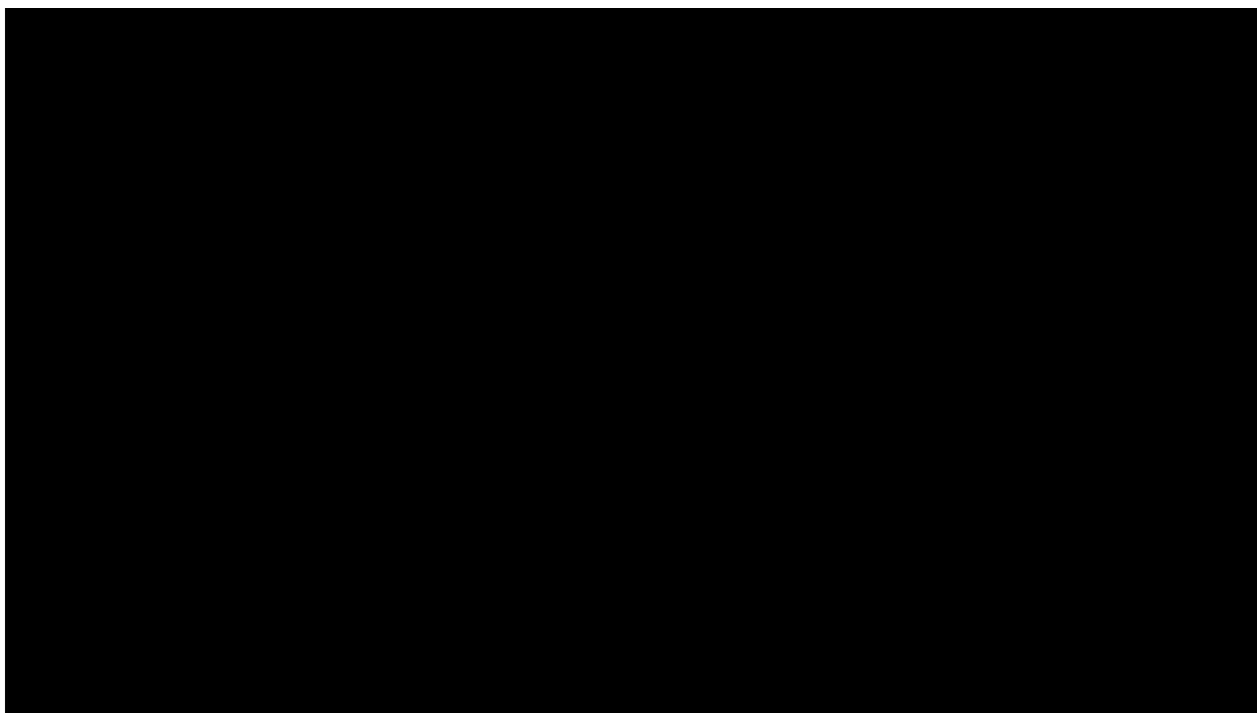
First entry: 14 Sep 2026 16:49:00 UTC
Latest entry: 14 Sep 2026 18:49:54 UTC

14 Sep 2026 16:49:00 | Candidate ready for review. Validation is incomplete or failed; inspect the recorded checks before using the patch.

14 Sep 2026 18:49:54 | Validation is incomplete or failed. Inspect the recorded gates.

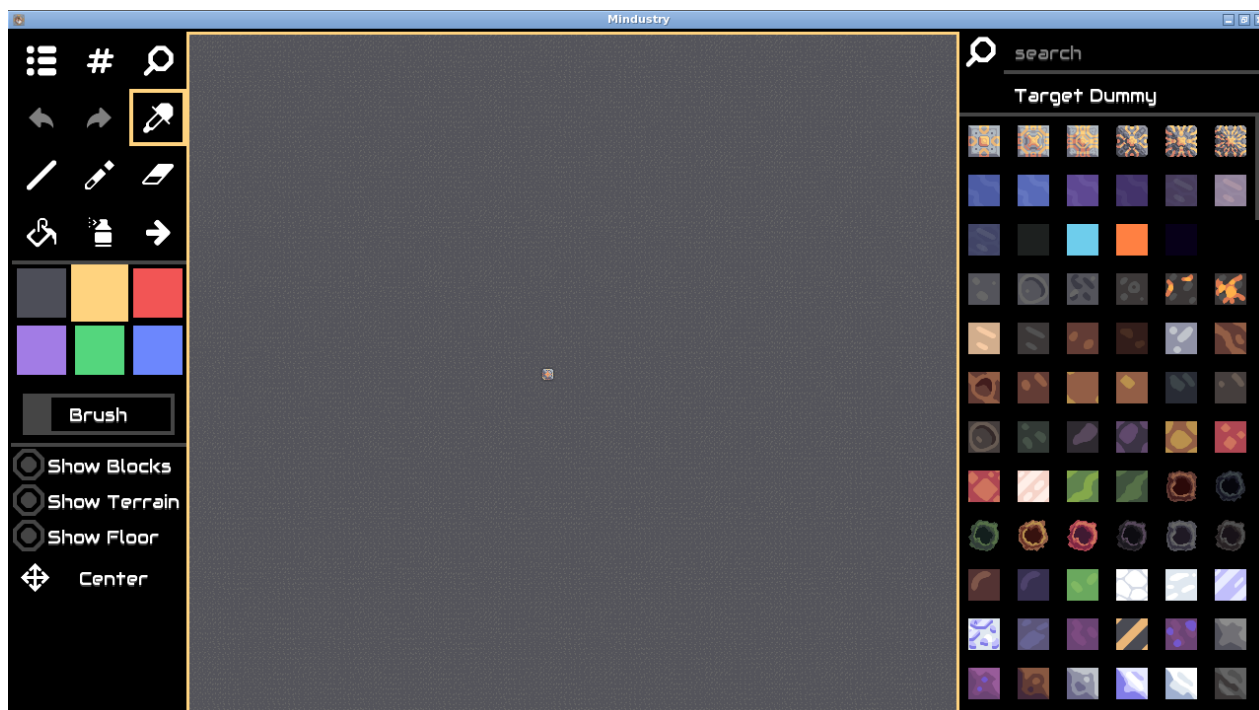
Recorded screenshots

Baseline replay evidence



md-12579-terra-overnight-01-2ab861941ebf8d998cb3-baseline-revalidation.png

Candidate replay outcome



md-12579-terra-overnight-01-52d95da04bb3c0170e2d-post-patch.png